How to report a dead network port

Article KB0010025 | Owner: Network Support | Subcategory: Ethernet / LAN | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 MacBook Pro/Air, USB-C/Ethernet docking stations, All Acme Global office sites (Bengaluru, Pune, Hyderabad, Mumbai, Chennai, Gurugram) and remote workers using VPN

Summary

This article guides users and service-desk agents through diagnosing and resolving a dead Ethernet wall-port issue at an Acme Global office location. It outlines quick self-help checks that most users can perform before opening a ticket, then provides the detailed troubleshooting workflow used by the Network Support team to restore connectivity.

Symptoms

- "Ethernet cable unplugged" notification in Windows network icon or macOS status bar
- Limited or no connectivity warning - yellow exclamation mark on the network adapter
- Unable to obtain an IP address; ipconfig shows media disconnected or 169.254.x.x APIPA address
- Ping to default gateway fails with "Request timed out."
- No link LED activity on the laptop NIC or docking-station LAN port

Common causes

- Faulty or damaged Ethernet patch cable (crushed RJ-45 connector, broken wires)
- Loose or improperly seated cable at the wall jack, patch panel, or switch port
- Defective wall-port jack or patch-panel termination (bent pins, corrosion)
- Switch port administratively shut down, error-disabled, or experiencing a hardware fault
- Out-of-date docking-station firmware or missing USB-C/Ethernet driver on the host OS
- Power-over-Ethernet (PoE) issues affecting devices that rely on PoE for link detection

Self-Help Checks and Basic Troubleshooting

Before contacting the Service Desk, users can perform a few quick checks that often restore a wired connection without needing a ticket. These steps take less than five minutes and require no special tools beyond the Ethernet cable already at the desk. If the problem persists after completing the procedure, gather the information listed in the "Information to Provide" bullet and open a ticket via the IT self-service portal, phone extension 4357, or the Teams "IT Help" channel.

1. 1. Verify physical connections
   - Ensure the Ethernet cable is firmly plugged into both the laptop (or docking-station LAN port) and the wall jack.
   - If you use a docking station, check that the USB-C/Ethernet adapter is securely attached to the dock and that the dock's own LAN LED shows activity.
   - Inspect the cable for visible damage (kinks, crushed connectors, exposed wires). If any damage is seen, replace the cable with a known-good spare.

2. Test with a different cable or port
   - Swap the current Ethernet cable for another one that you know works elsewhere (e.g., from a colleague's desk).
   - If a spare wall jack is available nearby, move the cable to that jack and see if the link comes up.
   - Note whether the link LED on the laptop NIC or dock flashes or stays solid.

3. Reset the network adapter
   - Windows: Open Settings -> Network & Internet -> Advanced network settings -> More network adapter options. Right-click the Ethernet adapter, choose Disable, wait 10 seconds, then Enable again.
   - macOS: Open System Settings -> Network, select Ethernet, click the gear icon -> Make Service Inactive, wait, then Make Service Active.
   - After toggling, wait up to 30 seconds for the adapter to obtain an IP address via DHCP.

4. Check IP configuration
   - Open a command prompt (Windows) or Terminal (macOS) and run `ipconfig /all` (Windows) or `ifconfig` / `ip addr show` (macOS).

- Look for an IPv4 address in the 10.x.x.x or 172.16.x.x ranges (Acme's internal DHCP scopes). If you see an APIPA address (169.254.x.x) or "Media disconnected," the link is still not established.

5. Restart the docking station (if applicable)
- Disconnect the dock from power and from the laptop for 10 seconds, then reconnect.
- Allow the dock to reboot; its Ethernet port should re-initialize and attempt to negotiate a link.

If after these steps you still have no link light, no IP address, or the "cable unplugged" message persists, proceed to open a ticket and provide the details below.

Information to Provide When Opening a Ticket

When you create a request, include the following specifics so the Network Support team can begin troubleshooting immediately. Providing this information reduces back-and-forth and helps meet the P2/P3 resolution targets.

1. - Exact location: building name, floor, desk or cubicle number, and wall-jack label (if visible).
- Device details: laptop make/model, docking-station make/model (if used), operating system version.
- Cable condition: note whether you inspected the cable and observed any damage, and whether you tried a spare cable.
- Test results: describe what happened when you moved the cable to a different jack or used a different cable (e.g., "link came up on jack B24", "no change with spare cable").
- Adapter state: output of `ipconfig /all` or `ifconfig` showing IP address, MAC address, and whether the media state is connected or disconnected.
- Any recent changes: new docking station, driver update, switch-port reconfiguration, or physical work near the desk.
- Screenshot or photo (optional) of the network icon showing the error, or of the cable connectors.

Service-Desk Procedure (Network Support)

Once a ticket is received, the Network Support analyst follows a structured workflow to isolate and resolve the dead port issue. The procedure assumes the user has already performed basic self-help checks; the analyst verifies those results and then proceeds to layer-2/layer-3 validation on the switch infrastructure.

1. 1. Ticket triage and information gathering
- Review the user-provided details (location, device, cable tests).
- Confirm the impact scope: is the issue isolated to a single desk, multiple adjacent ports, or a whole floor?
- Assign priority: typically P2 (High) if a single user loses wired connectivity; P1 only if the outage affects a critical server or many users.

2. Remote verification of link status
- Using the switch management interface (Palo Alto or Cisco switches as deployed), locate the switch port corresponding to the reported wall-jack (via patch-panel documentation or CDP/LLDP).
- Check the port status: look for `link down`, `err-disable`, or `admin down`. Note any error counters (CRC, collisions).
- If the port shows `link up` but the user reports no connectivity, verify VLAN assignment and DHCP snooping bindings.

3. Physical layer validation
- If the port is administratively up but link down, dispatch a field technician (or guide the user) to:
  a. Reseat the Ethernet cable at both ends.
  b. Replace the patch cable with a known-good spare.
  c. Inspect the wall-jack for bent pins, dust, or corrosion; clean or replace the jack if needed.
- For docking-station related cases, verify the dock's USB-C/Ethernet adapter firmware version; update if outdated per vendor guidance.

4. Switch port remediation
- If the port is err-disabled due to a security violation (e.g., port-security, BPDU guard), clear the err-disable state after confirming the cause is resolved.

- If the port is administratively shut down, enable it and monitor for link stability.
- Should the port persistently fail to link after cable and jack checks, schedule a port swap to a spare switch port and update patch-panel documentation.

5. Driver and firmware checks (host side)
- For Windows laptops, ensure the Ethernet controller driver is up to date via Company Portal or Windows Update.
- For macOS, verify that the latest Apple-provided driver for the USB-C/Ethernet chipset is installed (usually delivered through macOS updates).
- Request the user to reinstall the docking-station driver if the link only appears after a dock reboot.

6. Final validation and closure
- Ask the user to repeat the self-help verification steps (link LED, IP address, ping to default gateway).
- Confirm that the user can reach internal resources (e.g., file share `\fileserver eams`) and the internet.
- Document the root cause (cable fault, jack issue, switch port error, driver/firmware) and the corrective action taken.
- Close the ticket with a resolution note and advise the user to monitor the connection for the next 24?hours.

Verification and Closure

After the Network Support team believes the issue is resolved, both the analyst and the user should perform a quick validation to ensure the fix is stable before closing the ticket. This step helps avoid premature closure and reduces repeat incidents.

1. 1. Link-layer confirmation
- Verify that the Ethernet adapter reports a link speed of 1?Gbps (or 100?Mbps if that is the port's capability) and that the link LED is solid or blinking with activity.
- Run `ping <default-gateway>` for 30 seconds; look for 0?% loss and round-times under 5?ms.

2. IP-address and connectivity check
- Confirm that the device has received a DHCP lease from the correct scope (e.g., 10.10.20.0/24).
- Attempt to access an internal web application (e.g., the intranet home page) and an external site (https://www.acme.example) to validate both local and internet routing.

3. User feedback
- Ask the user to describe whether the original symptom ("cable unplugged" message, limited connectivity) has disappeared.
- If the user reports any intermittent loss, note the frequency and consider scheduling a port monitor or cable replacement.

4. Ticket closure
- Add a summary of the root cause and remedy to the ticket's resolution field.
- Set the status to Resolved and, after a 24-hour observation period with no further reports, move to Closed.
- If the issue recurs within that window, reopen the ticket and escalate to the senior Network Engineer for deeper investigation (possible cabling plant fault or switch hardware failure).

When to escalate

If the initial Network Support troubleshooting does not restore connectivity-e.g., the switch port continues to show link down after cable and jack replacement, or multiple adjacent ports exhibit the same symptom-escalate to the Senior Network Engineer within the Network Support group. Provide the following details in the escalation note: switch name/IP, affected port number, port status logs (showing err-disable reasons or link flaps), results of cable/jack tests, patch-panel mapping, and any recent change-management records (firmware upgrades, port re-configurations). The senior engineer will investigate potential switch-hardware faults, mis-configured port-security/VLAN settings, or broader cabling-plant issues and may schedule a site visit or initiate a hardware RMA if necessary.

Keywords: dead network port, ethernet not working, cable unplugged message, no network on docking station, limited connectivity, wired connection limited, desk network port dead, link down, err-disable, patch cable, wall jack, docking station Ethernet, USB-C Ethernet driver, switch port

troubleshooting, network support procedure